

Scalable ambient fabrication of single-junction and perovskite–silicon tandem solar cells

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Lingyuan Wang^{1,7}, Xiaodong Hu^{1,7}, Ben Niu², Yangyang Liu¹, Tianxiao Liu¹, Evan M. Wilson³, Zhen Jia⁴, Koucheng Chen⁴, Shumao Wang⁴, Zijia Li⁴✉, Yunjie Dou¹, Xiaoyu Shi¹, He Yan⁵✉, Qingdong Zheng⁶, Wei Wang²✉, Antonio Facchetti³✉ & Shangshang Chen¹✉

Transitioning high-efficiency perovskite solar cells from laboratory scale to industrial production requires scalable fabrication under ambient conditions without sacrificing performance or durability. Here we report ambient-air, blade-coated perovskite solar cells and modules that achieve power conversion efficiencies of 26.6% (26.5% certified) and 23.2%, respectively. These devices demonstrate exceptional stability, retaining 99.8% of their initial power conversion efficiency after 1,200 hours of continuous light soaking. This breakthrough is enabled by a hole-transporting co-polymer, PNCC, which synergistically integrates triarylamine and carbazole phosphonic acid units. Comprehensive spectroscopic, spectrometric and morphological analyses reveal the mechanisms underlying PNCC's superior functionality. Furthermore, leveraging PNCC's high conductivity and morphological uniformity, we demonstrate monolithic perovskite–silicon tandem cells with a certified efficiency of 33.0%. This work establishes a benchmark for ambient-processed, stable photovoltaics, providing a viable, scalable pathway towards sustainable solar energy harvesting.

Perovskite solar cells (PSCs) have risen as a next-generation photovoltaic technology for sustainable energy harvesting^{1–3}. Inverted PSCs with a positive–intrinsic–negative (p–i–n) architecture have gained considerable attention for their low-temperature processibility, greater device stability and compatibility with commercial silicon-bottom cells for tandem configurations^{4–9}. However, high-efficiency p–i–n PSCs are typically small devices fabricated by spin-coating in an inert atmosphere and require improved stability to meet industry

standards^{7,10–16}. For large-area production of p–i–n PSCs in ambient conditions, a critical step is coating a perovskite ink on the surface of a hole-transporting material (HTM) via blade or slot-die coatings¹⁷. In this process, the wettability, morphological uniformity and stability of the HTM on the transparent conductive oxide (TCO) substrate are critical parameters to ensure a high-quality perovskite film maximizing device performance^{18–23}. An ideal HTM should not only transport holes efficiently but also facilitate the coating and crystallization of the

¹State Key Laboratory of Coordination Chemistry, MOE Key Laboratory of High Performance Polymer Materials & Technology, School of Chemistry, Nanjing University, Nanjing, China. ²State Key Laboratory of Analytical Chemistry for Life Science, School of Chemistry, Nanjing University, Nanjing, China. ³School of Materials Science and Engineering, Georgia Institute of Technology, Atlanta, GA, USA. ⁴Chint New Energy Technology Co. LTD., Haining, China. ⁵Department of Chemistry, Hong Kong University of Science and Technology, Hong Kong, China. ⁶College of Engineering and Applied Sciences, Nanjing University, Nanjing, China. ⁷These authors contributed equally: Lingyuan Wang, Xiaodong Hu. ✉e-mail: zijia.li@Astronergy.com; hyan@ust.hk; wei.wang@nju.edu.cn; afacchetti@gatech.edu; schen@nju.edu.cn

perovskite film for scalable production. This is even more challenging for processing in ambient where environmental factors (for example, moisture level, oxygen) can substantially affect the perovskite film morphology and microstructure^{24–26}.

HTMs for p–i–n PSCs have undergone tremendous development, and two major organic HTM families are used in the state-of-the-art p–i–n PSCs: small-molecular self-assembled monolayers (SAMs), such as carbazole phosphonic acids (PACz), and polymeric materials, such as poly(triarylamine) (PTAA) and poly(3,4-ethylenedioxythiophene):poly(styrene sulfonate)^{27–30}. These HTMs have some advantages but also serious drawbacks. First, small-molecular PACz SAMs have a limited process window as well as poor uniformity, particularly on textured silicon substrates for tandem devices requiring conformal coverage, and thus they are unsuitable for mass production^{19,31}. Polymeric HTMs such as PTAA have better large-area uniformity and charge-transport ability^{32–34}, but the PTAA surface is too hydrophobic for reproducible large-area coating of perovskite inks³⁵. Poly(3,4-ethylenedioxythiophene):poly(styrene sulfonate) shows both excellent conductivity and wettability for perovskite inks, but it is very hygroscopic, and the strongly acidic $-\text{SO}_3\text{H}$ groups affect PSC stability³⁶. Recently, poly(carbazole phosphonic acid) HTM with improved conductivity and the anchoring effects of SAMs to TCO was developed, enabling improved solar cell efficiency and stability compared with carbazole SAMs¹⁹. However, prior studies have demonstrated that carbazole monomers exhibit a strong tendency to undergo self-cyclization rather than participating in cross-coupling reactions that lead to high-molecular-weight polymeric chains^{37–40}. Such deleterious cyclization reactions during homopolymerization of the carbazole monomer greatly limited its molecular weight and hole-transport ability. This limitation is intrinsic to the homopolymerization chemistry of carbazole derivatives, and a new design strategy is needed to implement this unit in polymeric HTMs.

Here we introduce PNCC, an HTM co-polymer synthesized via the co-polymerization of triarylamine and PACz units. This design suppresses carbazole cyclization, yielding high-molecular-weight polymers with exceptional hole transport. By integrating the high conductivity of PTAA with the robust indium tin oxide (ITO)-binding affinity of PACz, PNCC overcomes the poor wettability of PTAA and the structural uniformity of PACz. The resulting interfacial bonding with TCO substrates stabilizes the interfaces, passivates perovskite defects and optimizes crystallization, thereby substantially boosting device performance. Ambient-fabricated p–i–n PSCs utilizing PNCC achieve a maximum power conversion efficiency (PCE) of 26.6% (certified 26.5%) and retain 99.8% of their initial efficiency after 1,200 hours of light soaking at 40 °C. Further, PNCC enables large-area perovskite modules (19.11 cm² aperture area) to reach a 23.2% PCE and facilitates uniform deposition on textured silicon, delivering a certified 33.0% PCE in perovskite–silicon tandem cells. This HTM design provides a viable strategy for fabricating highly efficient, stable PSCs, advancing the commercialization of perovskite solar technology.

Results

Optoelectrical properties of HTMs

The PNCC co-polymer (Fig. 1a) was synthesized by co-polymerizing low-cost PACz and triarylamine units in a molar ratio of 1:3 under mild conditions. The incorporation of triarylamine monomers has dual functionality in enhancing co-polymer performance. First, these monomers effectively suppress undesirable cyclization of carbazole units through steric hindrance effects, thereby enabling higher degrees of polymerization. Second, their extended π -conjugated systems with extensive electron delocalization greatly enhance charge carrier mobility along the polymer backbone. This synergistic combination collectively is helpful to elevate the hole-transport capacity of the resulting co-polymer system. Consequently, the co-polymerization strategy improves the polymer molecular weight by about 5 $\times$ compared to the PACz homopolymer¹⁹. PNCC also exhibits good solubility in common

organic solvents, allowing facile solution processing. The PNCC thin films show a slightly wider bandgap of 3.0 eV (Supplementary Fig. 1) and a deeper valence band level of -5.35 eV (Supplementary Fig. 2) than PTAA, and thus it better matches that of methylammonium–formamidinium mixed-lead triiodide perovskites (-5.4 eV). PNCC-coated ITO substrates also exhibit smaller contact angles compared to counterparts based on PTAA and Me-4PACz (Supplementary Fig. 3), suggesting that the PNCC surface offers better wettability for common polar solvents used in perovskite inks, including *N,N*-dimethylformamide and 2-methoxyethanol (2-ME). The PNCC-coated ITO substrate also has the smallest surface roughness of 1.27 nm compared to those based on PTAA and Me-4PACz (Supplementary Fig. 4). The hole mobility of PNCC was accessed by space-charge-limited current measurements (Supplementary Fig. 5) and found to be greater ($5.4 \times 10^{-5} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) than those of PTAA ($2.2 \times 10^{-5} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) and Me-4PACz ($8.1 \times 10^{-6} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$). This is essential to effectively transport holes in relatively thick films, which allows for a wider device processing window for large-scale manufacturing. In contrast, PACz-type small-molecular HTMs suffer from a remarkable reduction in performance when their stacking exceeds a monolayer or few layers (1–3 nm) due to the relatively low hole mobility of PACz^{19,41}.

In addition, PNCC bonds to the ITO substrate via phosphonic acid groups. X-ray photo-electron spectroscopy (XPS) was employed to investigate the chemical interactions between the HTMs and ITO. As shown in Fig. 1b, the In 3d peaks of bare ITO shift towards high binding energy after its top is coated with Me-4PACz or PNCC, whereas no shifts were observed for the PTAA-covered ITO substrate. Additional carbon-phosphorus bond features assigned to PACz moieties were also observed (Supplementary Fig. 6). This confirms that PNCC has a much stronger binding to the underlying ITO than PTAA. These bondings of PNCC enable its strong adhesion to the underlying substrate, making it more resistant to light illumination and environmental stresses. The adhesion force of the HTMs to ITO was quantified by adhesion-force atomic force microscopy (AFM), and PNCC exhibits the highest adhesion force to the underlying ITO among these three HTMs (Supplementary Fig. 7). The enhanced adhesion of PNCC arises from a synergistic interplay between its functional binding motifs along the polymer backbone⁴² and the presence of undetached sites. These undetached sites effectively restrict the diffusion of detached motifs, promoting rebinding and increasing the overall adhesive strength. This mechanism allows for greater energy dissipation before rupture^{43,44}, contributing to the superior adhesive performance of PNCC. The multipoint anchoring property of PNCC enables its more robust binding to ITO and stabilizes the interface and devices under light soaking.

To characterize the deposition density of three HTMs on ITO, cyclic voltammetry measurements were carried out using HTM-coated ITO substrates as the working electrode according to the previously reported protocol^{45,46}. In this measurement, HTM-coated ITO substrates acted as the working electrodes and were swept at different scan rates. By fitting peak current with scan rates, the surface density of the HTMs can be derived (more measurement details can be found in the Supplementary Methods). As presented in Supplementary Fig. 8, PNCC exhibited a higher surface density compared to the Me-4PACz SAM, confirming the better coverage of PNCC blade-coated on ITO substrate. Photo-induced force microscopy was further used to examine the uniformity of three HTMs on ITO. The distributions of Me-4PACz and PNCC on ITO substrates were mapped at 1,300 cm⁻¹ (P=O bond vibration peak) and at 1,104 cm⁻¹ for the PTAA layer (C–N stretching peak). As presented in Supplementary Fig. 9, the Me-4PACz SAM exhibited notable non-uniformity and larger surface undulation in the absorption intensity, whereas both PTAA and PNCC HTMs exhibited more uniform distributions.

The crystallinity of perovskite films coated onto HTM-covered ITO substrates was investigated with X-ray diffraction analysis, and the perovskite film deposited on PNCC exhibited the highest diffraction

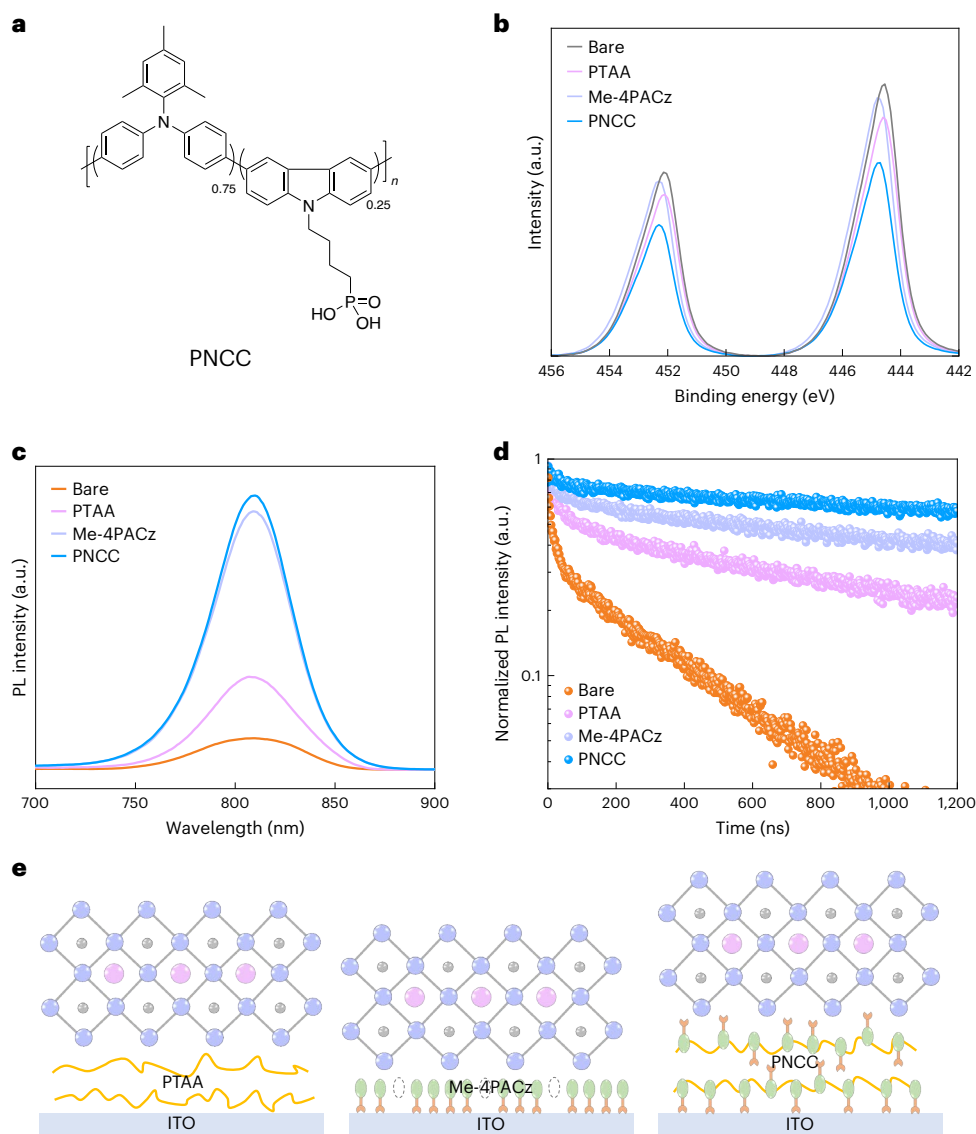


Fig. 1 | Characterizations of HTMs. **a**, Chemical structure of PNCC. **b**, XPS In 3d spectra of bare ITO and HTM-covered ITO substrates. All HTMs were coated with 1.0 mg ml^{-1} solutions to ensure a comparable thickness of $\sim 2 \text{ nm}$. **c**, **d**, PL (**c**) and TRPL (**d**) spectra of perovskites on glass substrates with or without HTMs. **e**, Schematic illustration of stacking three HTMs on ITO.

intensity, whereas no change in perovskite orientation or lattice parameters were observed (Supplementary Fig. 10). In addition, the buried PNCC/perovskite interface also showed fewer voids compared to the other samples (Supplementary Fig. 11). The effects of HTM coating on perovskite photoluminescence (PL) behaviour were also investigated by both steady-state and time-resolved photoluminescence (TRPL) characterizations, in which the excitation light was incident from the glass side. Compared with the bare $\text{MA}_{0.7}\text{FA}_{0.3}\text{PbI}_3$ perovskite film coated on a glass substrate, PNCC substantially enhances the PL intensity (Fig. 1c) and is much more effective in reducing non-radiative paths than both PTAA and Me-4PACz. In addition, the PL lifetime of the perovskite film on PNCC is longer than those on PTAA and Me-4PACz (Fig. 1d). To study the uniformity of PL emission, PL mapping of the perovskite films coated on three HTMs was characterized, and the perovskite deposited on PNCC also exhibited the highest PL intensity with improved emission uniformity (Supplementary Fig. 12). The enhanced PL intensity and prolonged PL lifetime can be attributed to the improved PNCC film quality and the passivation effects of phosphonic acid groups, suppressing non-radiative interfacial recombination. Building upon these results, Fig. 1e illustrates the proposed stacking

interaction of the three blade-coated HTMs on the ITO substrate. PTAA (6–7 nm) exhibits weak adhesion to ITO due to the lack of anchoring groups, and its inherent hydrophobicity hinders the formation of uniform perovskite films on top of it. Me-4PACz (1–3 nm), although possessing anchoring phosphonic acid groups that facilitate bonding with ITO, forms non-uniform films. In contrast, PNCC (9–10 nm, Supplementary Fig. 13) offers several advantages. It exhibits high conductivity and does not require an ultrathin layer for effective hole transport. Additionally, the phosphonic acid groups distributed throughout the thicker PNCC layer can still anchor to the ITO substrate and interact with the perovskite layer, enhancing interfacial stability and defect passivation. As shown in Supplementary Fig. 14, the Pb 4f XPS peaks of the buried PNCC/perovskite interface exhibited notable shifts after peeling off the ITO substrate, compared to those deposited on bare ITO.

Characterizations of single-junction perovskite solar devices

The photovoltaic performance of devices based on PNCC vis-a-vis PTAA and Me-4PACz HTMs was accessed by fabricating planar architectures of glass/ITO/HTM/ $\text{MA}_{0.7}\text{FA}_{0.3}\text{PbI}_3/\text{C}_{60}/\text{bathocuproine (BCP)}/\text{copper (Cu)}$ (Supplementary Fig. 15). Both HTM and perovskite layers were

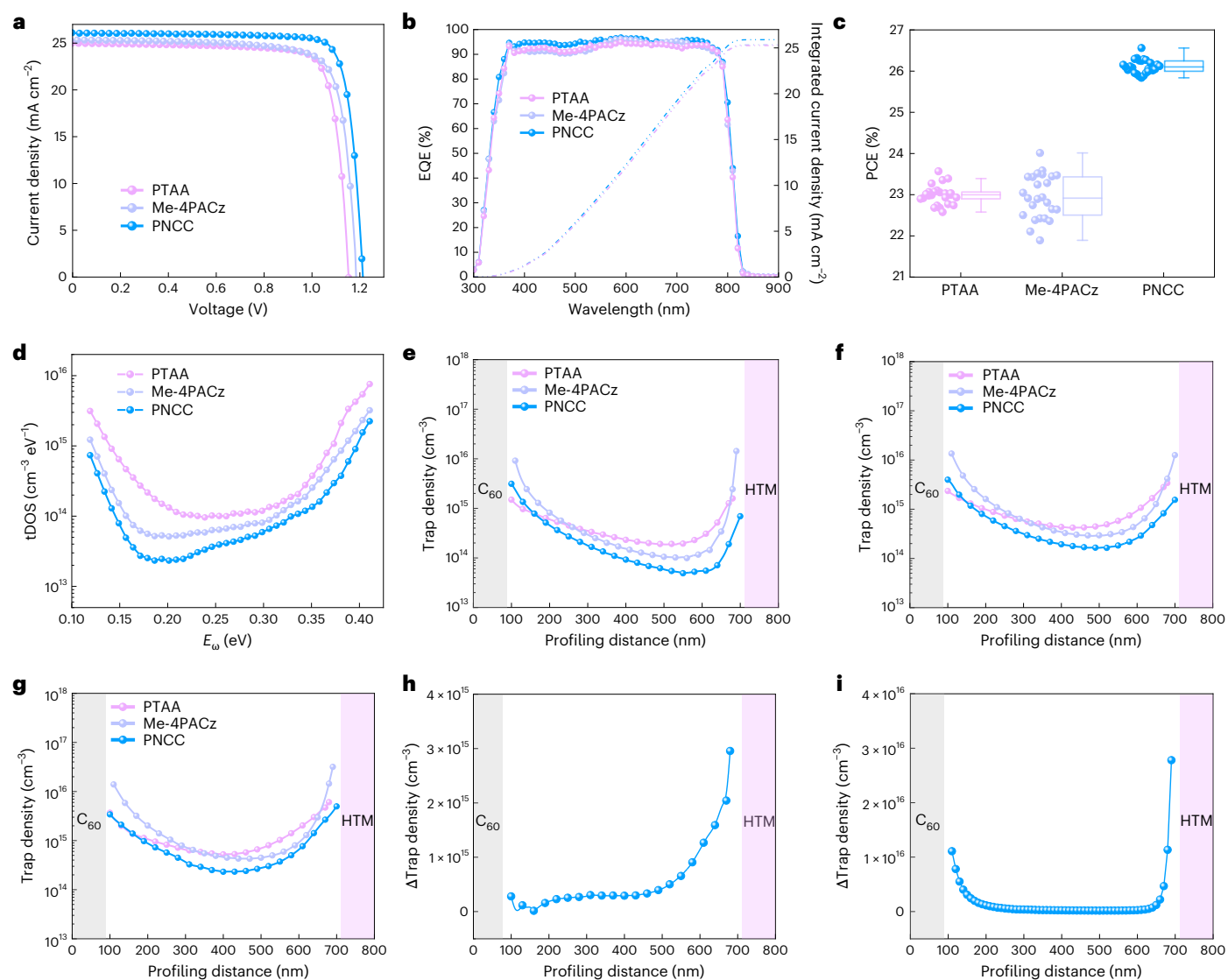


Fig. 2 | Photovoltaic performance of PSCs. a–d, Backward J - V characteristic curves (a), EQE spectra (b), PCE distributions (c) and tDOS spectra (d) of PSCs based on three HTMs. Box-plot elements (25 samples for each group) in c are defined as follows: centre line, median; box limits, upper and lower quartiles;

whiskers, $1.5\times$ interquartile range. **e–g,** DLCP trap profiling of the PSCs at frequencies of 200 kHz (e), 50 kHz (f) and 10 kHz (g). **h, i,** Changes in the deep trap density (10 kHz responding, linear scale) of PNCC cell with respect to PTAA (h) or Me-4PACz (i) cells. E_w , demarcation energy at a given angular frequency.

prepared by blade-coating in ambient conditions^{8,33}. The PSCs based on PNCC delivered a high PCE of 26.6% (Fig. 2a), along with an elevated open-circuit voltage (V_{oc}) of 1.21 V, a short-circuit current density (J_{sc}) of 26.1 mA cm^{-2} and a fill factor (FF) of 0.841, outperforming the reference cells based on PTAA (23.5% , V_{oc} : 1.15 V, J_{sc} : 25.0 mA cm^{-2} , FF: 0.819) and Me-4PACz (24.1% , V_{oc} : 1.19 V, J_{sc} : 25.3 mA cm^{-2} , FF: 0.800). Our champion cell based on PNCC was sent to an accredited third-party certification centre, and a PCE of 26.5% was confirmed (Supplementary Fig. 16), which represents an excellent value among air-processed PSCs irrespective of fabrication method or device structure and is comparable to those for devices fabricated in an inert atmosphere^{7,11}. The optical bandgap of the perovskite films is determined to be 1.51 eV (Supplementary Fig. 17), and thus the voltage deficit of our champion cell based on PNCC is merely 0.30 V. The champion PNCC cell also showed higher external quantum efficiency (EQE) response than the other two HTMs (Fig. 2b). The champion cell based on PNCC was further held at a fixed bias of 1.06 V, or maximum power point (MPP), and a stabilized power output for 300 s was recorded as depicted in Supplementary Fig. 18. The current density stabilizes at 24.7 mA cm^{-2} , thus corresponding to a stabilized PCE of 26.2%. In addition, the average PCEs of PNCC-based

cells are $26.1 \pm 0.2\%$, higher than those based on PTAA ($23.0 \pm 0.2\%$), Me-4PACz ($22.9 \pm 0.5\%$) (Fig. 2c and Supplementary Table 1) and also the reported Poly-4PACz ($24.3 \pm 0.7\%$, Supplementary Table 2). Such a statistical analysis corroborates the effectiveness of PNCC in improving the efficiencies of ambient-fabricated perovskite solar devices. In addition, the synthesis of PNCC has good reproducibility, and all six batches of PNCC were able to achieve PCEs over 26.0% (Supplementary Table 3).

To rationalize the high performance of PNCC-based cells, the trap density of state (tDOS) in these devices was assessed. The tDOS of three groups of PSCs were first studied by thermal admittance spectroscopy (Fig. 2d)^{47,48}. Clearly, the PNCC-based cell exhibits an overall lower trap density compared to its PTAA and Me-4PACz counterparts within the entire trap depth range, and the PTAA-based cell suffers from the highest trap density due to its weak HTM–ITO interaction and limited perovskite trap passivation. To further investigate the spatial reduction of trap states, we leveraged the drive-level capacitance profiling (DLCP) method to profile the spatial distributions of cell trap states (Supplementary Fig. 19)⁴⁹. As shown in Fig. 2e–g, the PNCC-based cell exhibits a considerable reduction in both shallow traps (high frequency responding, 200 kHz) and deep traps (low frequency responding,

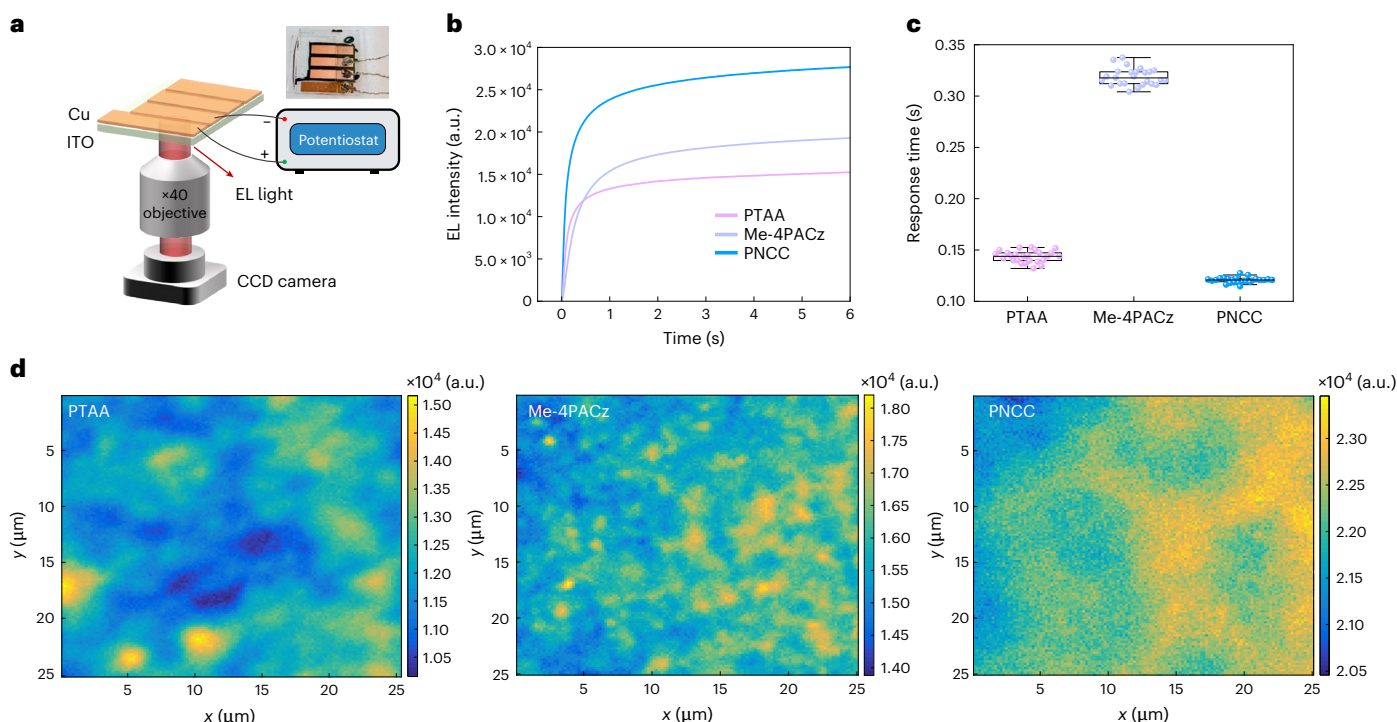


Fig. 3 | Spatial homogeneity investigation via EL mapping. **a**, Schematic illustration of microscopy coupled with a potentiostat to measure the EL of the PSCs in operando. **b**, EL intensity versus bias application time of three groups of PSCs. **c**, Summary of average EL response time determined from EL mappings

on 25 randomly selected zones. Box-plot elements are defined as follows: centre line, median; box limits, upper and lower quartiles; whiskers, 1.5× interquartile range. **d**, EL intensity mapping of the PSCs based on PTAA, Me-4PACz and PNCC. CCD, charge-coupled device.

10 and 50 kHz). It has been shown that the rich deep traps at the HTM/perovskite interface have critical effects in limiting the photovoltaic performance of p–i–n PSCs⁴⁹. We further plotted the change in deep trap density (10 kHz, $\text{density}_{\text{PTAA/Me-4PACz}} - \text{density}_{\text{PNCC}}$) to visualize the spatial trap-density difference among three types of PSCs (Fig. 2h,i). It is obvious that the deep trap density was reduced primarily near the HTM for the PNCC-based PSC. This corroborates the effectiveness of PNCC in passivating deep traps at the HTM/perovskite interface for devices processed in ambient conditions.

We also carried out electroluminescence (EL) mapping of the complete PSCs to investigate the microhomogeneity of blade-coated perovskite films on the three HTMs^{50,51}. Unlike conventional PL mapping, which provides insight only into the photoactive layers, EL mapping allows the characterization of a solar cell in a state close to its operational condition⁵². In addition, PL mapping typically suffers from interference from lateral carrier diffusion in the proximity of the excitation spot. Complementing PL mapping, EL mapping is able to quantitatively measure the luminescence flux of complete cells and was found to be free from various artifacts occurring from the analysis in a confocal configuration⁵². The schematic illustration of our EL mapping setup is shown in Fig. 3a, and a forward bias was applied to stimulate luminescence. The size of each EL detection zone is $25\ \mu\text{m} \times 25\ \mu\text{m}$, and the resolution of each pixel is 200 nm. Both EL intensity and response time can be derived in EL mapping. As plotted in Fig. 3b, the PNCC-based cell exhibited the fastest average EL rise and saturated at the highest EL intensity compared to the other types of cells. The fast EL rise for the PNCC cell is indicative of reduced interface traps that would negatively affect charge injection and radiative recombination. On the other hand, the high EL intensity is consistent with the suppressed non-radiative recombination, supporting the lowest voltage deficit of the PNCC PSCs. In addition, we also measured more randomly selected zones (25 zones for each group) of three groups of PSCs and plotted all of their average EL response time in Fig. 3c. Clearly, the PSCs based on PTAA and PNCC

exhibit much faster EL response times and narrower distributions than those of Me-4PACz, demonstrating the intrinsic advantages of polymeric HTMs versus SAMs for promoting charge transport. In addition, as shown in the EL maps (Fig. 3d and Supplementary Fig. 20), the PSC based on PNCC exhibits the best EL emission homogeneity; the Me-4PACz-based cells suffer from much poorer EL emission uniformity, as seen by the numerous dispersed luminous centres. These results corroborate the advantages of PNCC over conventional HTMs in terms of uniform deposition and fast charge transport. Overall, we concluded that the following factors contribute to these improvements based on our characterizations. First, PNCC shows much better uniformity on ITO and suppresses additional non-recombination paths at HTM/perovskite interface, as confirmed by the EL mapping, thus improving device V_{OC} . In addition, the strong interface interaction and high conductivity of PNCC reduce interface traps and hole accumulation, as shown in DLCP profiles and space-charge-limited current results, effectively transporting holes and improving cell J_{SC} .

Stability and scalability

We also examined the operational stability of three groups of encapsulated PSCs by connecting them to an automatic MPP tracker in air. With this setup, we could monitor the performance change of our cells at the MPP under a simulated one-sun illumination intensity of $100\ \text{mW cm}^{-2}$ (relative humidity 30–50%). No cooling or fans were used during the stability test, and the temperature of the cells was measured to be $-40\ ^\circ\text{C}$. As plotted in Fig. 4a, the PNCC-based cell exhibited high operational stability, with 99.8% of the initial PCE retained after 1,200 hours of light soaking. On the other hand, both PTAA- and Me-4PACz-based cells lost ~50% of their initial efficiency after ~1,000 hours of light soaking. Furthermore, we also compared the cell stability at elevated temperatures following the ISOS-L-2 standard protocol, and the PNCC-based cell retained 95.1% of initial PCE after 1,400 hours of light soaking at $65\ ^\circ\text{C}$ (Fig. 4b). Even at a harsh temperature of $85\ ^\circ\text{C}$, the PNCC-based

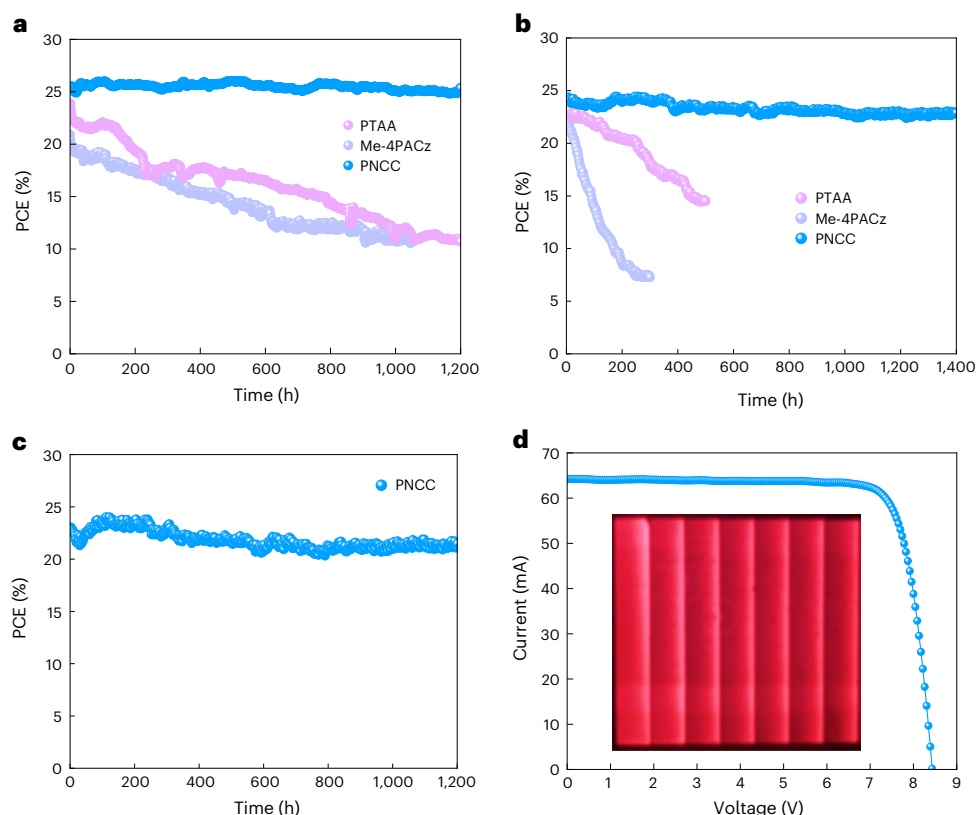


Fig. 4 | Stability and scalability of PNCC-based perovskite solar devices. **a–c**, MPP stability test results of encapsulated PSCs soaked under one-sun illumination in air at 40 °C (**a**), 65 °C (**b**) and 85 °C (**c**). **d**, J – V curve of champion

perovskite module based on PNCC. The inset shows a photograph of the PNCC-based perovskite module (aperture area: 19.11 cm²) and its EL emission under a forward bias.

cell was still able to maintain 92.8% of PCE after 1,200 hours of MPPT tracking (Fig. 4c). In addition, the ultraviolet (UV) stability of these devices was evaluated by soaking the cells under a 365-nm UV lamp with an intensity of 17.0 mW cm⁻², and their efficiency decays are plotted in Supplementary Fig. 21. Among these three groups, PNCC-based cells exhibited the best UV stability, whereas Me-4PACz-based cells suffered from the fastest degradation. This result is a notable advancement for air-processed PSCs, demonstrating high efficiency and superior stability. These results support that the PNCC HTM is crucial in achieving stable p–i–n PSCs. The improved PNCC-based device stability is attributed to, among other factors, the strong bonding of the phosphonic acid groups of PNCC to ITO.

We further evaluated PNCC's compatibility with the upscaling process by fabricating perovskite solar modules in ambient conditions. These modules realized an impressive PCE of 23.2% at an aperture area of 19.11 cm², along with a V_{oc} of 8.44 V, a short-circuit current (I_{sc}) of 64.2 mA and an FF of 0.816 (Fig. 4d). The geometry fill factor of the module was 94.7% (Supplementary Fig. 22), corresponding to an active-area (18.15 cm²) efficiency of 24.5%. The EL technique also enabled us to examine the macro-uniformity of the large-area solar module⁵³. Under a forward bias of -9.3 V, the PNCC-based module delivered bright photoemission from the entire active area without obvious defective regions (inset in Fig. 4d), which demonstrated the excellent uniformity of the perovskite film on PNCC.

Perovskite–silicon tandem solar cells

The improved conductivity and outstanding uniformity of PNCC make it a highly promising candidate for two-terminal perovskite–silicon tandem solar cells. A critical challenge in such architectures lies in achieving conformal HTM deposition on the textured pyramid structures

of silicon-bottom cells (Fig. 5a). Conventional SAMs deposited via scalable methods often struggle to achieve conformal coverage due to their inherent non-uniformity⁵⁴, necessitating the current industrial workaround of a double-layer sputtered NiO_x/SAM—a solution that undermines cost-effectiveness and scalability⁵⁵. Here we explore the application of PNCC as a standalone HTM for perovskite–silicon tandem devices, in which a wide-bandgap Cs_{0.05}Rb_{0.05}FA_{0.75}MA_{0.15}Pb(I_{0.75}Br_{0.25})₃ (1.69-eV bandgap) perovskite works as the front subcell (Supplementary Fig. 23). As illustrated in Fig. 5b, PNCC demonstrated exceptional surface conformity on textured silicon subcells without detectable aggregates, whereas notable aggregates were observed after coating PTAA or Me-4PACz on the top (Supplementary Fig. 24). This superior interfacial quality directly translated to enhanced device performance: PNCC-based tandem solar cells yielded a remarkable PCE of 33.3% (Fig. 5c), along with a high V_{oc} of 1.99 V (independently certified as 33.0%; Supplementary Fig. 25). The EQE spectra of two subcells matched each other well (Supplementary Fig. 26). Statistical analysis of 18 tandem solar cells is presented in Supplementary Fig. 27 and Supplementary Table 4 with an average PCE of (32.0 ± 0.5)%. Such an efficiency outperformed that based on NiO_x/Me-4PACz (31.2%; Supplementary Fig. 28). These results not only validate PNCC's compatibility with industrially relevant textured silicon but also highlight its potential to streamline the fabrication of high-efficiency tandem photovoltaics by eliminating complex multilayer HTM architectures. Meanwhile, we also found that PNCC was able to suppress the phase segregation in wide-bandgap perovskite (Supplementary Fig. 29), which is beneficial for cell stability. The operational stability of the encapsulated tandem solar cell was also studied, and the perovskite–silicon cell was able to retain 85% of its initial efficiency after 1,140 hours of one-sun illuminance at 65 °C in air (Supplementary Fig. 30).

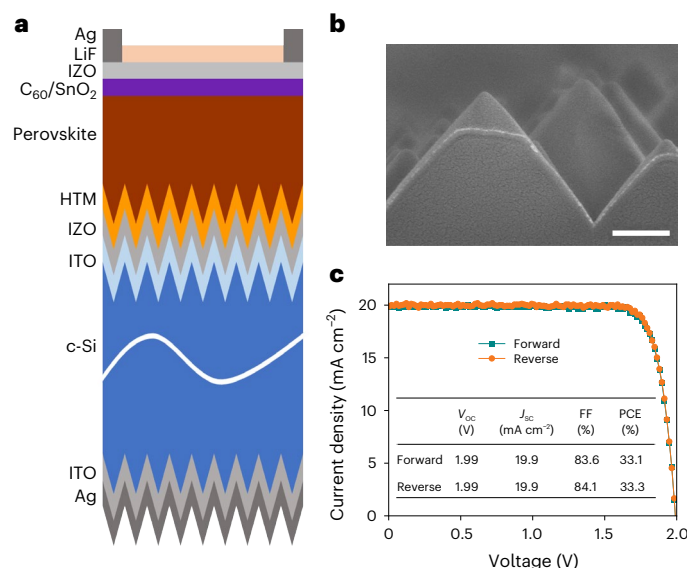


Fig. 5 | Perovskite tandem solar cells. **a**, Schematic illustration of the perovskite tandem device structure. **b**, SEM image of the PNCC layer on textured silicon half cell (scale bar: 500 nm). **c**, J - V curves of the champion perovskite tandem cell based on PNCC. Inset shows the detailed photovoltaic parameters.

Discussion

We design an HTM via co-polymerization of PACz with triarylamine units. The newly developed HTM enables robust anchoring to the perovskite TCO surface while maintaining efficient charge-transport capability and high-quality film uniformity in perovskite solar devices. PSCs have emerged as a promising photovoltaic technology for sustainable solar energy harvesting, offering great potential to reduce greenhouse gas emissions and advance environmental sustainability. Nevertheless, the commercialization of perovskite photovoltaics remains hindered by two critical bottlenecks: scalability and operational stability. In this work, we address these key challenges through rational charge-transport material design and demonstrate scalable and stable single-junction and tandem perovskite solar devices fabricated in ambient conditions. Our findings provide a feasible strategy to accelerate the commercialization progress and further enhance the environmental sustainability of perovskite photovoltaic technology.

Methods

Materials

All chemicals without any notes were commercially available and used without further purification. PTAA (average M_n 7,000–10,000), 2-ME, BCP, lead iodide (PbI_2 , 99.999% trace metals), dimethyl sulfoxide, L - α -phosphatidylcholine (LP), benzylhydrazine hydrochloride, and toluene were purchased from Sigma-Aldrich and used without further purification. Me-4PACz was purchased from TCI Chemicals. Methylammonium iodide, formamidinium iodide, 4-fluoro-phenethylammonium iodide, n-dodecylammonium iodide and other perovskite precursor salts were purchased from GreatCell Solar. C_{60} was purchased from Shenzhen eFlex Inc.

Fabrication of single-junction perovskite solar devices

Patterned ITO glass substrates were first cleaned by ultrasonication with soap, deionized water and isopropyl alcohol and then UV-ozone treated for 15 min before use. All perovskite solar devices were prepared by blade-coating at room temperature inside a fume hood with a relative humidity of $40 \pm 10\%$. Me-4PACz was blade-coated onto ITO glass substrates with a concentration of 1 mg ml^{-1} dissolved in methanol at a speed of 20 mm s^{-1} , and the gap between the blade-coater (BEVS Industrial Co., LTD, model 1811GV) and ITO

substrates was $150 \mu\text{m}$. PTAA with a concentration of 3.3 mg ml^{-1} dissolved in toluene was blade-coated onto ITO glass substrates at a speed of 20 mm s^{-1} . PNCC with a concentration of $2\text{--}5 \text{ mg ml}^{-1}$ dissolved in chloroform:methanol (3:1, v/v) was blade-coated onto ITO glass substrates at a speed of 20 mm s^{-1} . Me-4PACz and PNCC layers were thermally annealed at 100°C for 10 min in air, and no thermal annealing was necessary for the PTAA layer. The thicknesses of PTAA and PNCC layers were measured to be 6–7 nm and 9–10 nm by profilometry (ZEPTOOLS JS100A), respectively. A $1.30 \text{ M MA}_{0.7}\text{FA}_{0.3}\text{PbI}_3$ precursor solution was prepared by dissolving the corresponding organic halide salts and lead iodide into 2-ME in a N_2 -filled glove box with an O_2 level $< 10 \text{ ppm}$. Then 0.83 mg ml^{-1} n-dodecylammonium iodide, 0.27 mg ml^{-1} LP, 0.14 v/v MAH_2PO_2 , 1.40 mg ml^{-1} 4-fluoro-phenethylammonium iodide, 0.15 mg ml^{-1} benzylhydrazine hydrochloride and $2.8\% \text{ v/v}$ dimethyl sulfoxide were added as additives before blade-coating. Subsequently, the precursor solution was taken out of the glove box and blade-coated onto the HTM-covered ITO glass substrates with a gap of $250 \mu\text{m}$ at a movement speed of 20 mm s^{-1} in air. The air knife worked at 20 psi during blade-coating. After that, the perovskite films were annealed at 120°C for 10 min in air to remove the residual solvents. The thickness of the perovskite films was $\sim 800 \text{ nm}$. The solar cells were completed by thermally evaporating C_{60} (25 nm , $0.3 \text{ \AA s}^{-1}$), BCP (5 nm , $0.1 \text{ \AA s}^{-1}$) and 100 nm copper ($1 \text{ \AA s}^{-1}$) with a high-vacuum evaporator purchased from Shenyang Kecheng Vacuum Tech Co., LTD (model ZD-Z400). The active areas of the solar cells were $8 \text{ mm} \times 2 \text{ mm}$, determined by a metal shadow mask). The PSCs were encapsulated with a cover glass sealed by epoxy encapsulant (Gorilla 2-part epoxy 4200130) on the back. For mini-modules, laser scribing was performed twice with a Keyence laser marker (MD-U1000C, 355 nm). The laser power for P2 and P3 scribing was 0.375 W at a scribing speed of 500 mm s^{-1} . The total scribing line width was 0.32 mm , giving a geometry fill factor of 94.7% , and each subcell had a total area of 2.73 cm^2 ($0.60 \text{ cm} \times 4.55 \text{ cm}$). The polydimethylsiloxane (PDMS) antireflection layer was prepared by a soft lithography method, and a textured silicon wafer with a pyramid size of $\sim 10 \mu\text{m}$ was used as a template. First, the PDMS prepolymer/curing agent mixture (SYLGARD 184 Silicone Elastomer, 10:1 w/w) was dropped onto the silicon wafer and allowed to rest for 0.5 h to allow the PDMS to fill the cavities. After being cured on a hot plate kept at 150°C for 15 min, the PDMS layer was peeled off the silicon wafer with caution and then mounted onto the surface of the module substrate with adhesive tape.

Fabrication of perovskite tandem solar cells

Silicon heterojunction-bottom solar cells were prepared using czochralski n-type silicon wafers with a nominal thickness of $110 \mu\text{m}$. A double-sided submicron texture structure was obtained by soaking the wafers in alkaline solution. After a standard cleaning and hydrofluoric acid dip process to remove the silicon oxide layer, a hydrogenated intrinsic amorphous (i-a-Si:H) layer with a thickness of approximately 5–6 nm was grown on both sides of the czochralski wafers by plasma-enhanced chemical vapour deposition. Afterwards, hydrogenated phosphorus-doped nanocrystalline silicon oxide ($\text{nc-SiO}_x\text{:H}$) and hydrogenated boron-doped nanocrystalline silicon (nc-Si:H) layers were deposited on the front and back sides, respectively. Then, 80-nm ITO and 10-nm indium zinc oxide (IZO) were sputtered on both sides of the cell as the rear electrode and the recombination layer, and the $1.1 \times 1.1 \text{ cm}^2$ square cell was defined by a metal mask. The wafers were then annealed at 180°C for 10 min under 20-sun illumination to recover the sputtering damage. Ag was thermally evaporated on the rear surface, and the silicon-bottom cells were laser-cut to $2.5 \times 2.5 \text{ cm}^2$ for tandem preparation. For the perovskite top cells, the substrates were treated in a UV-ozone cleaner for 5 min. The PNCC layer was coated on the IZO layer and then annealed at 100°C for 10 min. Then, the $1.6 \text{ M Cs}_{0.05}\text{Rb}_{0.05}\text{MA}_{0.15}\text{FA}_{0.75}\text{Pb}(\text{Br}_{0.25}\text{I}_{0.75})_3$ perovskite precursor solution was

coated and annealed for 10 min at 100 °C. After that, the C₆₀ (15 nm) was evaporated. SnO₂ (12 nm) was deposited by atomic layer deposition with N₂ as the gas carrier. Tetrakis(dimethylamino)tin(IV) was used as the Sn precursor, and water was used as oxidant. The substrate temperature was maintained at 85 °C. The dosing and purging time for Tetrakis(dimethylamino)tin(IV) was 0.5 and 5 s and for H₂O was 0.5 and 5 s. One hundred cycles were performed. After atomic layer deposition of SnO₂, the IZO top electrode (50 nm) was deposited on the buffer layer by direct-current sputtering in the Ar/O₂ gas mixture (~6% vol O₂) using a power of 2 kW. Finally, 1 µm of patterned Ag and 110 nm of LiF were thermally evaporated on top of the IZO.

Device and material characterizations

The *J*–*V* characteristics of solar cells and modules were assessed using a 3-A LED solar simulator (BG-LED3A-100S, Class AAA Solar Simulator) in air at room temperature, and the intensity of the simulated light was calibrated to 100 mW cm⁻² by a silicon reference cell (Newport 91150V). All solar devices were measured using a Keithley 2400 source metre with a scan rate of 0.1 V s⁻¹, and the delay time was 10 ms. There was no preconditioning before measurement. A metal mask with an aperture (7.4 mm²) aligned with the single-junction solar cell area was used for measurements, and the aperture areas of the mask for tandem cells was 0.9814 cm². UV–vis absorption spectra were obtained with a Shanghai Metash UV-8000 spectrometer. UV photoelectron spectroscopy spectra were acquired with a UV photo-electron spectrometer (Surface Analysis Kratos Axis Ultra DLD) under a vacuum of 2 × 10⁻⁸ mbar. The incident photon energy (He I) was 21.22 eV, and a -10 V bias was applied on the samples to enhance the signals with kinetic energy near zero. XPS was performed using a Thermo Scientific NEXSA facility with an Al K α anode source (1,486.6 eV) operated at 15 kV and 10⁻⁹ mbar chamber pressure. Photo-induced force microscopy images were obtained with a neaspec neaSCOPE (Molecular Vista) microscope equipped with an infrared (IR) laser, which has a gap-free narrowband tunable wavenumber range of 550–2,250 cm⁻¹. The HTMs were blade-coated on ITO substrates following the same procedure as for single-junction cells. The sample was measured with fixed wavenumbers (1,300 cm⁻¹ for PNCC and Me-4PACz, 1,104 cm⁻¹ for PTAA), and the scan speed was set at 1.9 µm s⁻¹. As for adhesion-force AFM measurement, an SNL-10 cantilever (Type A, from Bruker) was used, and a polystyrene sphere (20 µm in diameter) was attached to the cantilever using epoxy. The force measurements were conducted on a commercial AFM (NanoWizard IV from JPK). The HTM-coated ITO glass sheet was glued onto a glass slide for subsequent AFM measurements. We randomly selected three areas on the sample surface for adhesion-force measurement: in each area, a 20 × 20 matrix of data points was set up, with force measurements performed at each data point. During a typical AFM measurement, the polystyrene sphere modified cantilever was approached to the sample surface at a constant speed of 2 µm s⁻¹ until the target setpoint of 0.5 nN was reached. The cantilever was held at 0.5 nN on the sample surface for 2 seconds before being retracted at the same speed. The force-extension curves were recorded and analysed using commercial software (JPK Data Processing). EQE spectra were obtained with a Zolix SSC600 EQE system equipped with a standard Si cell. The stability of the encapsulated PSCs was monitored with a 91 PVK SOLAR MSCLT-1 automatic MPP tracker (one-sun illumination intensity of 100 mW cm⁻²), and the cells kept working at MPP conditions in air (30–50% relative humidity). tDOS and DLCP spectra were obtained with an Agilent E4980A precision LC metre. Both steady-state PL and TRPL spectra were acquired on a HORIBA FL-3 fluorescence spectrophotometer at room temperature, and the perovskite films were blade-coated onto the glass substrates with or without HTMs. The excitation wavelength for steady-state PL measurements was 405 nm with an intensity of ~70 mW cm⁻² coming from a Xenon lamp. The excitation laser used in TRPL measurements was produced by Horiba DeltaDiode-L 405 nm with an intensity of

63 mW cm⁻². The excitation light was incident from the glass side for both PL and TRPL measurements. The PL intensity mapping images were obtained in a scanning area of 10 × 10 µm² by the ISS Q2 laser confocal microscopy system with a ×60 water immersion microscope objective lens; the perovskite was excited with a 405-nm laser, and the emission was filtered by a 652-nm long-pass filter. The film thickness was determined by a stylus profiler (ZEPTOOLS JS100A). Scanning electron microscopy images were taken on a Regulus 8100 field-emission scanning electron microscope. AFM height images were obtained with a Bruker Icon atomic force microscope. The routine IR spectrum of PNCC powder was acquired with a Bruker FT-IR Tensor II Spectrometer. The imaging system of EL mapping was conducted using an inverted microscope (Eclipse Ti-U, Nikon) with an Auto-lab potentiostat (PGSTAT302N). A forward voltage was applied to stimulate luminescence. The EL light from the cell was captured by a charge-coupled device camera (Stingray, Allied Vision Technologies) through a ×40 objective (NA = 0.6). A data acquisition card (USB-6281, National Instruments) was employed to synchronize the voltage output from the potentiostat and transistor–transistor logic signals from the camera. ¹H NMR spectra were obtained on a 500-MHz Bruker AVANCE III-500 spectrometer. High-resolution mass spectrometry was performed using a Thermo Fisher Q Exactive LC/MS.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data generated or analysed in this study are provided in the published article and its Supplementary Information. Source data are provided with this paper.

References

- Chen, Y. et al. Driving a green energy transition with halide perovskite solar cells. *Nat. Sustain.* <https://doi.org/10.1038/s41893-026-01844-5> (2026).
- Tian, X. et al. Perspectives for sustainability analysis of scalable perovskite photovoltaics. *Energy Environ. Sci.* **18**, 194–213 (2025).
- Goetz, K. P., Taylor, A. D., Hofstetter, Y. J. & Vaynzof, Y. Sustainability in perovskite solar cells. *ACS Appl. Mater. Interfaces* **13**, 1–17 (2021).
- Zhu, R. Inverted devices are catching up. *Nat. Energy* **5**, 123–124 (2020).
- Zhu, P. et al. Toward the commercialization of perovskite solar modules. *Adv. Mater.* **36**, 2307357 (2024).
- Chen, P. et al. Multifunctional ytterbium oxide buffer for perovskite solar cells. *Nature* **625**, 516–522 (2024).
- Li, Z. et al. Stabilized hole-selective layer for high-performance inverted p-i-n perovskite solar cells. *Science* **382**, 284–289 (2023).
- Chen, S. et al. Stabilizing perovskite-substrate interfaces for high-performance perovskite modules. *Science* **373**, 902–907 (2021).
- Li, X. et al. Constructing heterojunctions by surface sulfidation for efficient inverted perovskite solar cells. *Science* **375**, 434–437 (2022).
- Liu, C. et al. Bimolecularly passivated interface enables efficient and stable inverted perovskite solar cells. *Science* **382**, 810–815 (2023).
- Liang, Z. et al. Homogenizing out-of-plane cation composition in perovskite solar cells. *Nature* **624**, 557–563 (2023).
- Bai, S. et al. Planar perovskite solar cells with long-term stability using ionic liquid additives. *Nature* **571**, 245–250 (2019).
- Chen, R. et al. Reduction of bulk and surface defects in inverted methylammonium- and bromide-free formamidinium perovskite solar cells. *Nat. Energy* **8**, 839–849 (2023).

14. Li, H. et al. 2D/3D heterojunction engineering at the buried interface towards high-performance inverted methylammonium-free perovskite solar cells. *Nat. Energy* **8**, 946–955 (2023).
15. Park, S. M. et al. Engineering ligand reactivity enables high-temperature operation of stable perovskite solar cells. *Science* **381**, 209–215 (2023).
16. Feng, K. et al. Non-fullerene electron-transporting materials for high-performance and stable perovskite solar cells. *Nat. Mater.* **24**, 770–777 (2025).
17. Li, Z. et al. Scalable fabrication of perovskite solar cells. *Nat. Rev. Mater.* **3**, 18017 (2018).
18. Howard, I. A. et al. Coated and printed perovskites for photovoltaic applications. *Adv. Mater.* **31**, 1806702 (2019).
19. Ren, Z. et al. Poly(carbazole phosphonic acid) as a versatile hole-transporting material for p-i-n perovskite solar cells and modules. *Joule* **7**, 2894–2904 (2023).
20. Zheng, X. P. et al. Co-deposition of hole-selective contact and absorber for improving the processability of perovskite solar cells. *Nat. Energy* **8**, 462–472 (2023).
21. Xu, W. et al. Multifunctional entinostat enhances the mechanical robustness and efficiency of flexible perovskite solar cells and minimodules. *Nat. Photon.* **18**, 379 (2024).
22. Li, B. et al. Fundamental understanding of stability for halide perovskite photovoltaics: the importance of interfaces. *Chem* **10**, 35–47 (2024).
23. He, R. et al. Improving interface quality for 1-cm² all-perovskite tandem solar cells. *Nature* **618**, 80–86 (2023).
24. Chen, S., Xiao, X., Gu, H. & Huang, J. Iodine reduction for reproducible and high performance perovskite solar cells and modules. *Sci. Adv.* **7**, eaab8130 (2021).
25. Yan, L. et al. Fabrication of perovskite solar cells in ambient air by blocking perovskite hydration with guanabenz acetate salt. *Nat. Energy* **8**, 1158–1167 (2023).
26. Liu, K. et al. Moisture-triggered fast crystallization enables efficient and stable perovskite solar cells. *Nat. Commun.* **13**, 4891 (2022).
27. Magomedov, A. et al. Self-assembled hole transporting monolayer for highly efficient perovskite solar cells. *Adv. Energy Mater.* **8**, 1801892 (2018).
28. Puerto Galvis, C. E., González Ruiz, D. A., Martínez-Ferrero, E. & Palomares, E. Challenges in the design and synthesis of self-assembling molecules as selective contacts in perovskite solar cells. *Chem. Sci.* **15**, 1534–1556 (2024).
29. Zhang, C. et al. A review on organic hole transport materials for perovskite solar cells: structure, composition and reliability. *Mater. Today* **67**, 518–547 (2023).
30. Duan, Y. et al. A comprehensive review of organic hole-transporting materials for highly efficient and stable inverted perovskite solar cells. *Adv. Funct. Mater.* **34**, 2315604 (2024).
31. Phung, N. et al. Enhanced self-assembled monolayer surface coverage by ALD NiO in p-i-n perovskite solar cells. *ACS Appl. Mater. Interfaces* **14**, 2166–2176 (2022).
32. Deng, Y. et al. Defect compensation in formamidinium-caesium perovskites for highly efficient solar mini-modules with improved photostability. *Nat. Energy* **6**, 633–641 (2021).
33. Deng, Y. et al. Tailoring solvent coordination for high-speed, room-temperature blading of perovskite photovoltaic films. *Sci. Adv.* **5**, eaax7537 (2019).
34. Kothandaraman, R. K. et al. Sputtered NiO interlayer for improved self-assembled monolayer coverage and pin-hole free perovskite coating for scalable near-infrared-transparent perovskite and 4-terminal all-thin-film tandem modules. *Sol. RRL* **8**, 2400176 (2024).
35. Degani, M. et al. 23.7% efficient inverted perovskite solar cells by dual interfacial modification. *Sci. Adv.* **7**, eabj7930 (2021).
36. Boldyreva, A. G. et al. Unraveling the impact of hole transport materials on photostability of perovskite films and p-i-n solar cells. *ACS Appl. Mater. Interfaces* **12**, 19161–19173 (2020).
37. Coumont, L. S. & Veinot, J. G. C. Ni(COD)₂ coupling of 3,6-dibromocarbazoles as a route to all-carbazole shape persistent macrocycles. *Tetrahedron Lett.* **56**, 5595–5598 (2015).
38. Ostrauskaite, J. & Strohhriegl, P. Formation of macrocycles in the synthesis of poly(N-(2-ethylhexyl)carbazol-3,6-diyl). *Macromol. Chem. Phys.* **204**, 1713–1718 (2003).
39. Zhu, H. et al. Cyclization-promoted ultralong low-temperature phosphorescence via boosting intersystem crossing. *J. Am. Chem. Soc.* **143**, 2164–2169 (2021).
40. Zhu, H. et al. Cyclo[4]carbazole, an iodide anion macrocyclic receptor. *Org. Lett.* **18**, 5054–5057 (2016).
41. Zhang, S. et al. Minimizing buried interfacial defects for efficient inverted perovskite solar cells. *Science* **380**, 404–409 (2023).
42. Fan, H. et al. Adjacent cationic–aromatic sequences yield strong electrostatic adhesion of hydrogels in seawater. *Nat. Commun.* **10**, 5127 (2019).
43. Li, Y. et al. Molecular design principles of lysine-DOPA wet adhesion. *Nat. Commun.* **11**, 3895 (2020).
44. Yang, J., Bai, R., Chen, B. & Suo, Z. Hydrogel adhesion: a supramolecular synergy of chemistry, topology, and mechanics. *Adv. Funct. Mater.* **30**, 1901693 (2020).
45. Park, S. M. et al. Low-loss contacts on textured substrates for inverted perovskite solar cells. *Nature* **624**, 289–294 (2023).
46. Liu, M. et al. Compact hole-selective self-assembled monolayers enabled by disassembling micelles in solution for efficient perovskite solar cells. *Adv. Mater.* **35**, 2304415 (2023).
47. Shao, Y., Xiao, Z., Bi, C., Yuan, Y. & Huang, J. Origin and elimination of photocurrent hysteresis by fullerene passivation in CH₃NH₃PbI₃ planar heterojunction solar cells. *Nat. Commun.* **5**, 5784–5790 (2014).
48. Bao, C. & Gao, F. Physics of defects in metal halide perovskites. *Rep. Prog. Phys.* **85**, 096501 (2022).
49. Ni, Z. et al. Resolving spatial and energetic distributions of trap states in metal halide perovskite solar cells. *Science* **367**, 1352–1358 (2020).
50. Mochizuki, T. et al. Solar-cell radiance standard for absolute electroluminescence measurements and open-circuit voltage mapping of silicon solar modules. *J. Appl. Phys.* **119**, 034501 (2016).
51. Chaturvedi, P., Hoex, B. & Walsh, T. M. Broken metal fingers in silicon wafer solar cells and PV modules. *Sol. Energy Mater. Sol. Cells* **108**, 78–81 (2013).
52. El-Hajje, G. et al. Quantification of spatial inhomogeneity in perovskite solar cells by hyperspectral luminescence imaging. *Energy Environ. Sci.* **9**, 2286–2294 (2016).
53. Deng, Y. et al. Reduced self-doping of perovskites induced by short annealing for efficient solar modules. *Joule* **4**, 1949–1960 (2020).
54. Kore, B. P. et al. Efficient fully textured perovskite silicon tandems with thermally evaporated hole transporting materials. *Energy Environ. Sci.* **18**, 354–366 (2025).
55. Luo, X. et al. Efficient perovskite/silicon tandem solar cells on industrially compatible textured silicon. *Adv. Mater.* **35**, 2207883 (2023).

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Author contributions

S.C. conceived and designed the research. L.W. synthesized PNCC. X.H. fabricated and measured the PSCs. B.N. carried out EL characterization. Y.L. performed laser scribing. T.L. fabricated the perovskite modules. E.M.W. assisted with the material characterization. Z.J., K.C. and S.W. fabricated the perovskite-silicon tandem solar cells. Y.D. carried out the characterizations related to trap states. X.S. assisted with the device fabrication. Q.Z. supervised the optoelectronic characterization. Z.L. supervised the tandem cell fabrication. W.W. supervised EL characterization. S.C., H.Y. and A.F. wrote the manuscript with inputs from all co-authors. All authors discussed the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Zijia Li, He Yan, Wei Wang, Antonio Facchetti or Shangshang Chen.

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Research sample	Hole-transporting materials and their perovskite solar devices were used for this study. Details can be found in Methods section.
Sampling strategy	Sampling procedures were provided in the Results and discussion, and Methods section of the main text.
Data collection	The data were collected using standard practice in this type of work. Details can be found in the Methods section.
Timing and spatial scale	The samples were collected at predetermined times, which have given in the Methods of the main text.
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